

Word Embeddings

```
!pip install gensim scikit-learn matplotlib
from gensim.models import Word2Vec
from nltk.tokenize import word_tokenize
import nltk
```

```
nltk.download('punkt')
nltk.download('punkt_tab') # Added to download the missing resource
```

```
# Sample corpus
corpus = [
    "Natural language processing is a fascinating field",
    "Word embeddings capture semantic meanings",
    "NLP is used in chatbots and virtual assistants",
    "Word2Vec is a powerful tool for creating word embeddings"
]
```

```
# Tokenize sentences
tokenized_corpus = [word_tokenize(sentence.lower()) for sentence in corpus]
print(tokenized_corpus)
```

Output

```
[[['natural', 'language', 'processing', 'is', 'a', 'fascinating', 'field'], ['word', 'embeddings', 'capture', 'semantic', 'meanings'], ['nlp', 'is', 'used', 'in', 'chatbots', 'and', 'virtual', 'assistants'], ['word2vec', 'is', 'a', 'powerful', 'tool', 'for', 'creating', 'word', 'embeddings']]]
```

```
# Train Word2Vec model
model = Word2Vec(sentences=tokenized_corpus, vector_size=100, window=5, min_count=1, workers=4)
```

```
# Save the model
model.save("word2vec.model")
```

```
model = Word2Vec.load("word2vec.model")
```

```
# Get vectors for a subset of words
words = list(model.wv.index_to_key)[:10] # Select the first 10 words
print(words)
word_vectors = [model.wv[word] for word in words]
```

```
print(word_vectors)
```

Output

```
['is', 'embeddings', 'word', 'a', 'creating', 'for', 'tool', 'powerful', 'word2vec', 'assistants']
```

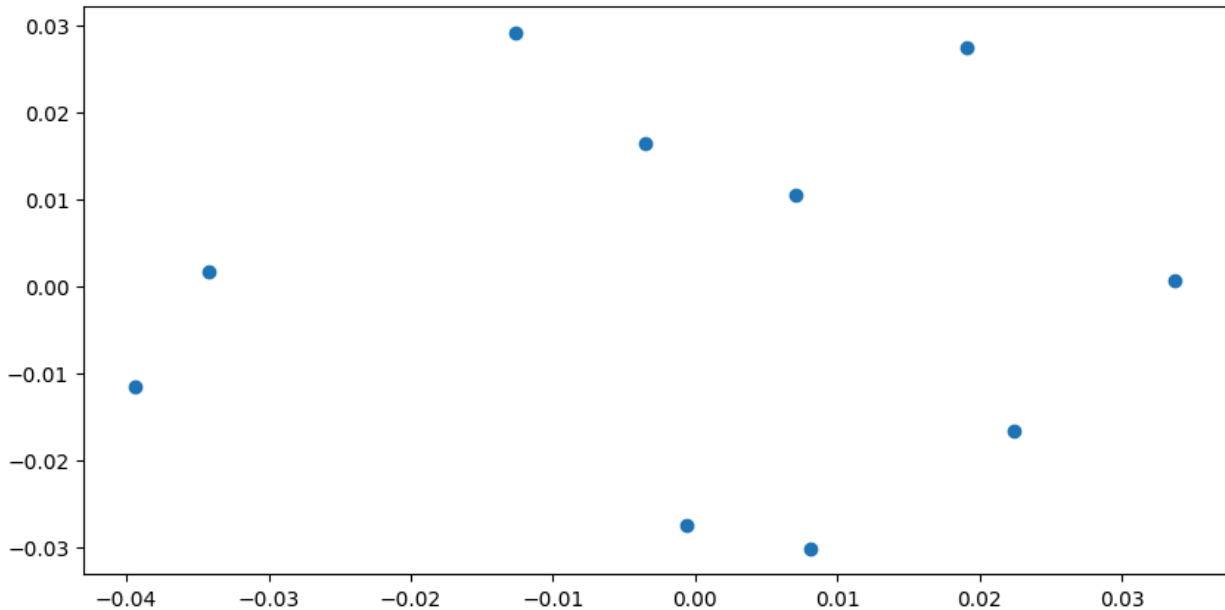
```
from sklearn.decomposition import PCA
```

```
# Apply PCA for dimensionality reduction  
pca = PCA(n_components=2)  
pca_result = pca.fit_transform(word_vectors)
```

```
import matplotlib.pyplot as plt
```

```
# Plot the words in 2D space  
plt.figure(figsize=(10, 5))  
plt.scatter(pca_result[:, 0], pca_result[:, 1])
```

Output



```
# Annotate the points with the words  
for i, word in enumerate(words):  
    plt.annotate(word, xy=(pca_result[i, 0], pca_result[i, 1]))
```

```
plt.title("2D Visualization of Word Embeddings")
plt.xlabel("PCA Component 1")
plt.ylabel("PCA Component 2")
plt.grid(True)
plt.show()
```

